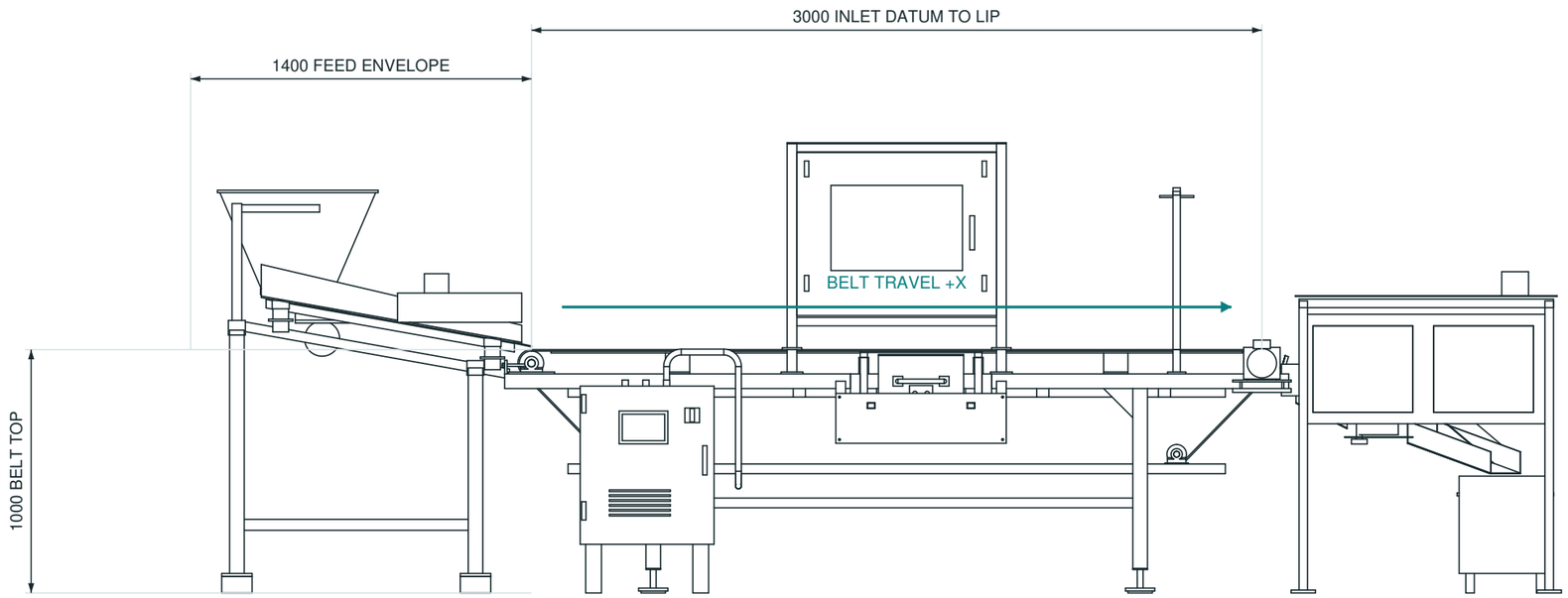
# General arrangement

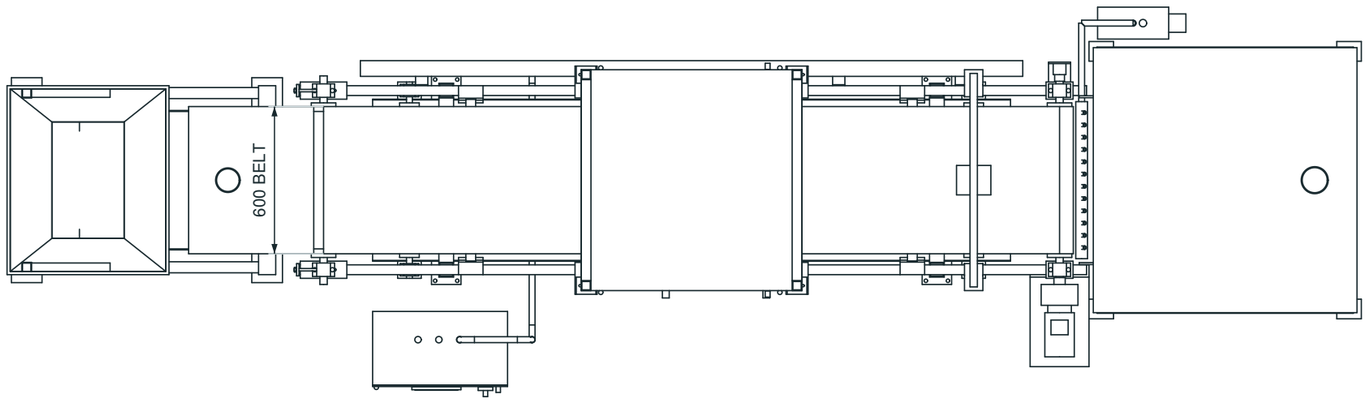
Proposed PCB-rich recovery module / depolluted, liberated, dry screened WEEE / geometry only






**SIDE / actual assembly, covers retained / scale per drawing / 1:25**

Side scale 1:25; nominal floor z=0. Feed and discharge hardware are concept selections.




**PLAN / same solid assembly, orthographic +Z view / 1:25**

## DESIGN INTENT

One airjet target/balance split; a separate balance-path gate diverts conservative review windows.

## COORDINATE DATUMS

x follows belt travel; y is transverse; z is height above floor. Inlet x=0 and lip x=3000 are fixed coordination datums.

| Station | x (mm) |
|---|---|
| RGB centre | 1300 |
| Induction centre | 1600 |
| Last active datum | 1750 |
| Cassette edge | 1775 |
| Discharge lip | 3000 |

## KEEP THESE DISTINCT

The sensing datum is 25 mm before the enclosure edge. The 1250 mm tracking distance starts at x=1750, not x=1775.

## DRAWING STATUS

STEP-derived views show packaging and mechanical intent. Strength, wear, drive sizing, guarding and component selection remain open.

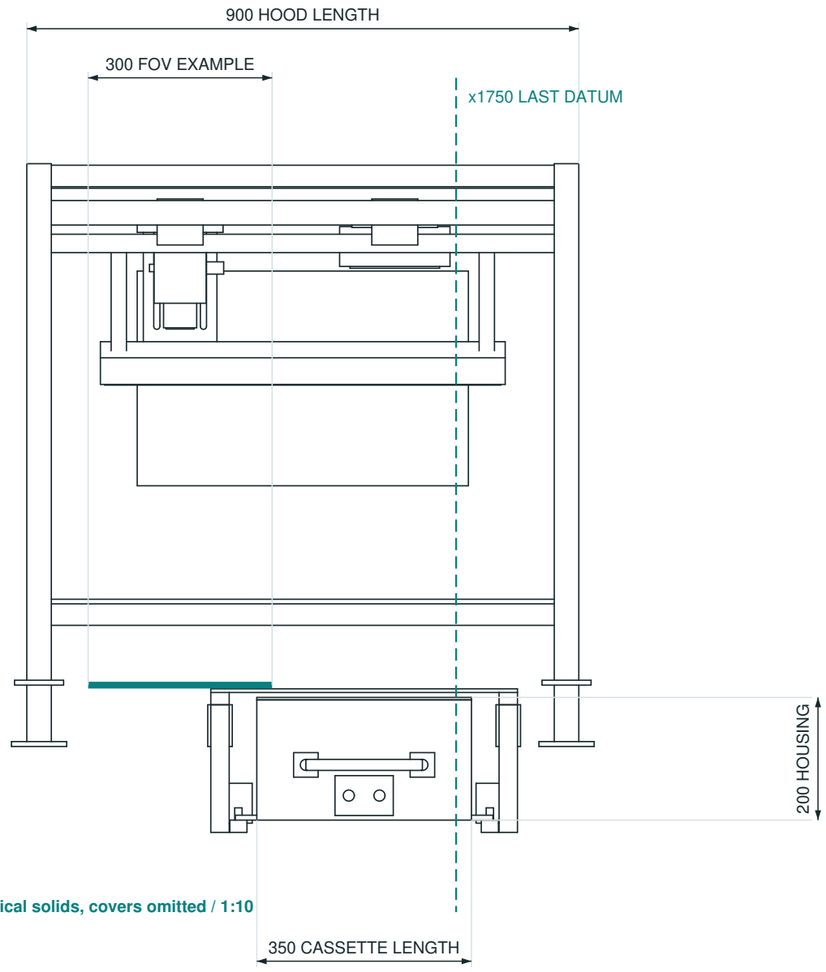
**DIMENSIONS: mm | SCALE: PER VIEW AT A3 | DO NOT SCALE FOR MANUFACTURE**

Selected layout geometry. No tolerances, load rating, sensing performance or safety compliance is established.

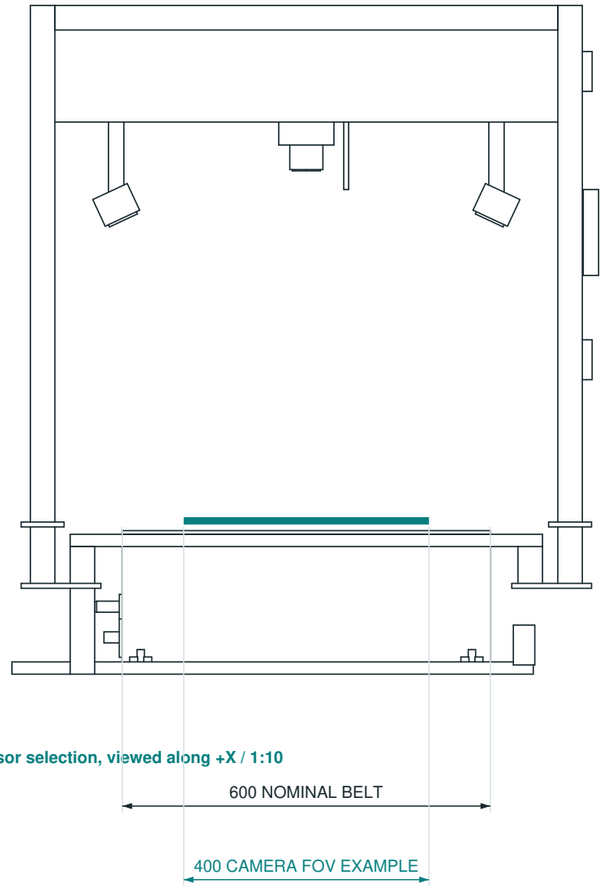
Source: conveyor_assembly.step | SHA-256 c4b5179c836f98c5 | 380 part instances | basis 2026-10-13

# Sensing cell and coverage

Adjustable optical and induction envelopes / cutaway selection omits opaque hood panels




**SENSOR SIDE / selected physical solids, covers omitted / 1:10**




**TRANSVERSE VIEW / sensor selection, viewed along +X / 1:10**

## CONDITIONAL CASSETTE REMOVAL

Remove the left access panel, withdraw locating pins and disconnect connectors. Reserve an engineered isolation procedure. This drawing checks clearance only, not safe servicing.

## LIFT-OFF IS NOT THIS HOUSING GAP

Selected housing-to-window gap: 8 mm, packaging only. Coil plane, belt/window effects and object shape require calibration.

## COVERAGE REQUIRES CALIBRATION

400 mm optical coverage is an example, not full 600 mm belt coverage. The separate 500 mm effective-width calculation is an assumption. Usable sensing width is TBD.

**DIMENSIONS: mm | SCALE: PER VIEW AT A3 | DO NOT SCALE FOR MANUFACTURE**

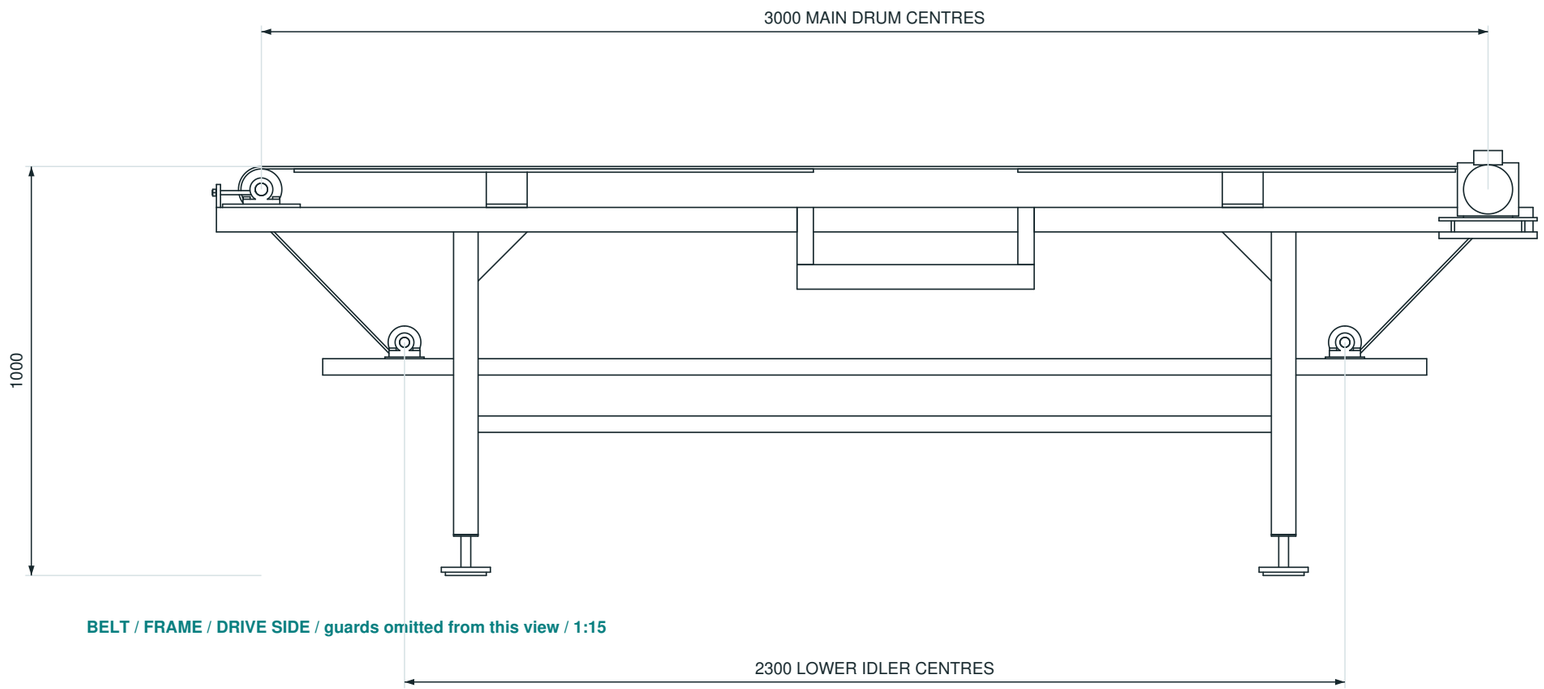
Selected layout geometry. No tolerances, load rating, sensing performance or safety compliance is established.

Source: conveyor_assembly.step | SHA-256 c4b5179c836f98c5 | 380 part instances | basis 2026-10-13

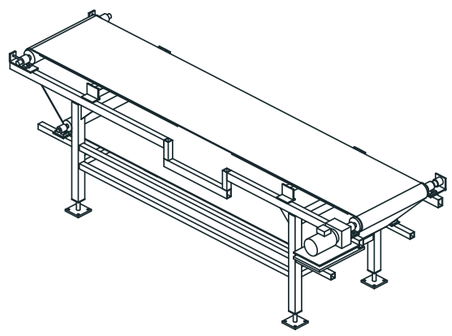
# Belt path, support and drive

Selected mechanical layout / covers removed for visibility / sizes are not strength-checked






**BELT / FRAME / DRIVE SIDE / guards omitted from this view / 1:15**



**BASE ISOMETRIC / actual CAD solids / no manufacturing scale**

## BELT PATH SELECTION

Main drums: diameter 100, centres z=944. Belt thickness 6. Lower idlers: diameter 80, centres z=570. Lower belt outer bottom z=524.

## SENSOR KEEP-OUT

The lowered return run clears the induction housing at z=780..980. The nonconductive window spans x=1350..1850; placeholder thickness 6 requires deflection and wear tests.

## DRIVE AND TENSION

Motor, gearbox, bearings, shafts, tension screws and mounts are modeled packaging selections. Motor torque, belt tension, tracking authority and bearing life are not calculated.

## RELEASE INPUTS

Define maximum piece mass, impact/load cases, belt construction and joint, roller crown, shaft/bearing fits, drive duty and restart torque before detailing fabrication.

## PROTECTION

Guard and access-panel geometry is a concept allowance. Nip clearances, interlocks, isolation, emergency functions and dust controls require a competent machinery review.

**DIMENSIONS: mm | SCALE: PER VIEW AT A3 | DO NOT SCALE FOR MANUFACTURE**

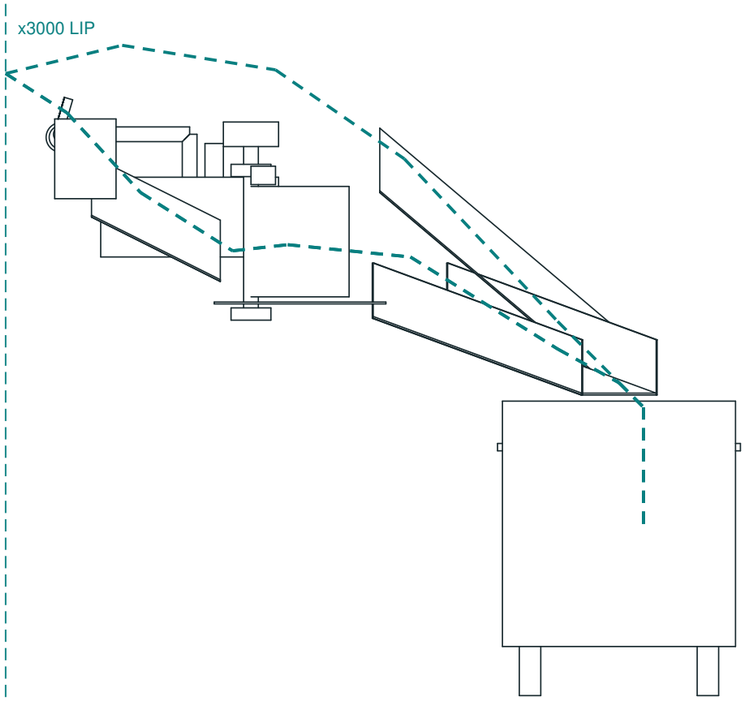
Selected layout geometry. No tolerances, load rating, sensing performance or safety compliance is established.

Source: conveyor_assembly.step | SHA-256 c4b5179c836f98c5 | 380 part instances | basis 2026-10-13

# Discharge and review routing

One proposed airjet split plus downstream balance-window gate / no precise three-way sorting claim



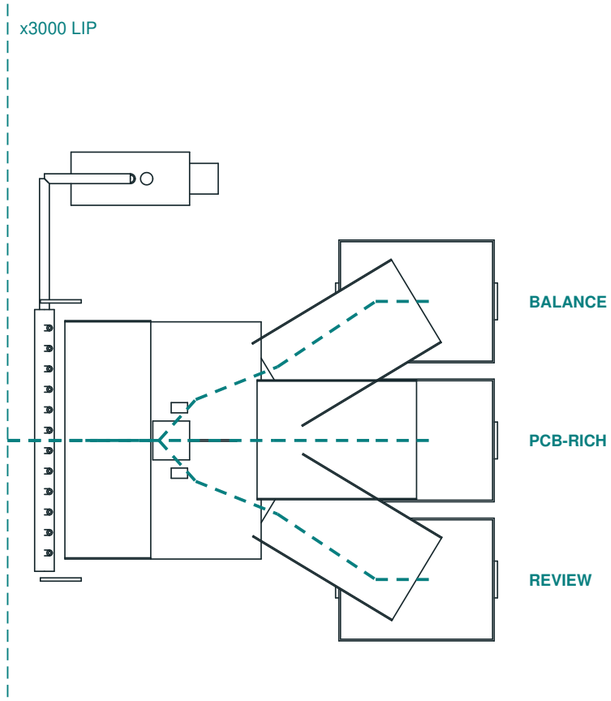



**DISCHARGE SIDE / selected solids, protective shell omitted / 1:10**

Dashed lines: route intent only; not simulated or measured flight paths.

### TARGET PATH

The primary air bar is proposed to deflect PCB candidates into concentrate. Nozzle pitch, pressure, impulse, timing and splitter position are unvalidated. Actual trajectories require representative trials.




**DISCHARGE PLAN / physical chute and collection layout / 1:15**

### BALANCE / REVIEW PATH

Gate shown in transition position: no feed permitted. It acts on balance only. Inhibit feed or verify a clear gap, allow a conservative drain window, command and verify position, then resume.

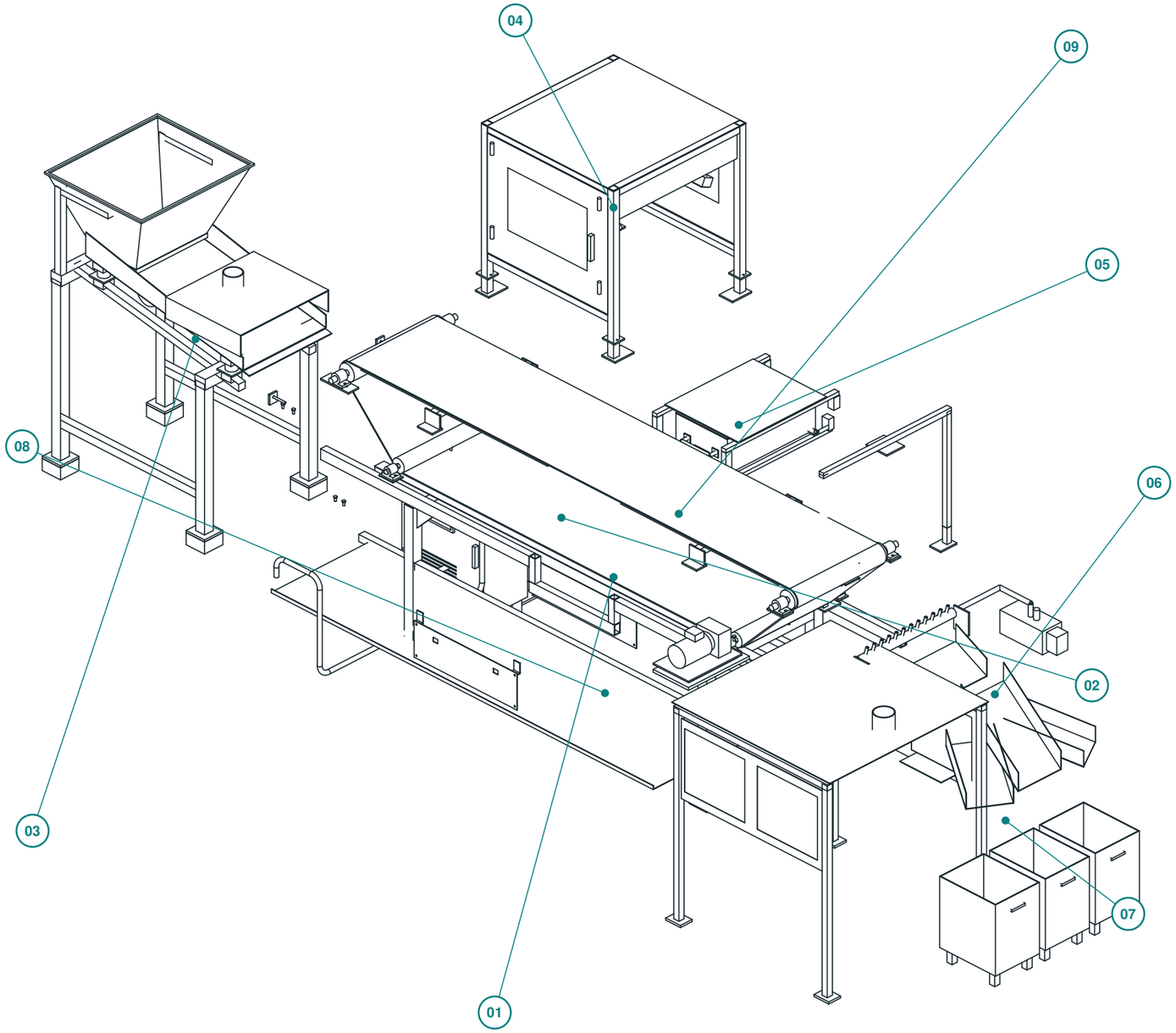
### ROUTING BOUNDARY

Review routing may include neighbouring pieces and cannot recover an unknown already ejected into concentrate. Chute bounce prevents exact item-time assumptions. Feed inhibit and a normal review bin are not certified safety functions or battery containment.

# Assembly groups and concept BOM

Exploded arrangement from the same STEP solids / group translation for visibility only






**EXPLODED ISOMETRIC / arbitrary separation / not to scale**

| No. | Assembly group | Parts |
|---|---|---|
| 01 | Base, deck and belt | 59 |
| 02 | Drive / bearings / fixings | 48 |
| 03 | Feeder / spreader | 59 |
| 04 | Hood, light, camera | 71 |
| 05 | Sensor and window | 27 |
| 06 | Airjet and chutes | 31 |
| 07 | Gate and bins | 29 |
| 08 | Guards and cable | 32 |
| 09 | Controls / research | 24 |

**380 modeled part instances; quantities from manifest.**

## BOM INTERPRETATION

Counts describe modeled solid part instances, not purchased assemblies. Sensor, actuator and cabinet internals are envelope models. Fixings shown are illustrative; final fastener schedules remain open.

## SOURCE TRACEABILITY

The part manifest retains names, categories, bounding boxes and solid volumes. The parametric CadQuery source is the editable design definition. STEP is the checked exchange solid assembly.

## NOT A PROCUREMENT RELEASE

Material grades, finishes, fabrication details, approved vendors, quantities of unmodeled internals, torque values and acceptance tolerances are not specified.

**DIMENSIONS: mm | SCALE: PER VIEW AT A3 | DO NOT SCALE FOR MANUFACTURE**

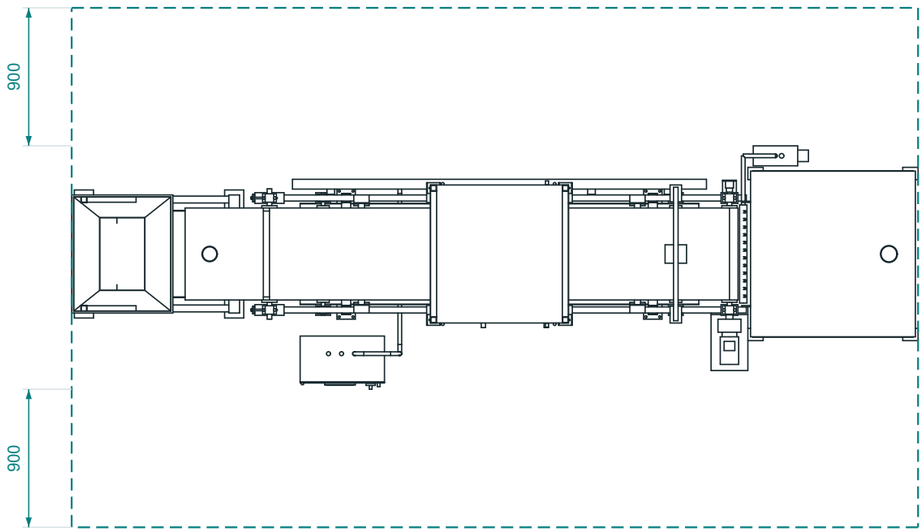
Selected layout geometry. No tolerances, load rating, sensing performance or safety compliance is established.

Source: conveyor_assembly.step | SHA-256 c4b5179c836f98c5 | 380 part instances | basis 2026-10-13

# Installation envelope and trial gates

Layout reservations, ideal kinematics and tests still needed / no rated machine performance






**PLAN / 900 mm side zones from outer equipment / 1:40**

## SERVICE SPACE IS A RESERVATION

Measure side zones from the outer hardware envelope, including projecting cabinet or drive parts. Door, cassette-withdrawal and bin-exchange sweeps need separate site checks. End access, utilities and actual dust-control connections remain to be designed.

## FEED BOUNDARY

Only depolluted, liberated, dry screened WEEE. Run 12.5-25 and 25-50 mm campaigns separately. Exclude batteries, whole devices, hazards, tangles and oversized/heavy pieces; sieve size does not bound length, thickness or mass.

## IDEAL KINEMATICS

| m/s | mm/frame* | s to lip** |
|---|---|---|
| 0.5 | 16.7 | 2.5 |
| 1 | 33.3 | 1.25 |
| 2 | 66.7 | 0.625 |

* At 30 fps reference. ** 1250 mm / speed. Assumes steady belt speed, no slip or collision. These are calculated values, not latency or throughput evidence.

## PROPOSED TEST RAMP

50-200 kg/h is an experimental ramp, not rated capacity. Begin slow and sparse at 0.5 m/s; 1 m/s is the calculation reference and 2 m/s is an unvalidated study case.

## MEASURE BEFORE CLAIMING

Calibrated sensor coverage and lift-off; presentation/overlap; synchronized tracking/latency; ejection scatter and neighbours; chute dwell and gate feedback; balanced mass, purity, recovery and actual buyer value.

## SEPARATE EXISTING EVIDENCE

The recorded steel/copper YOLO demo remains a separate artifact. It is not PCB inference, physical recovery or evidence of this concept's profitability.